

Aditya Balaji

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Quantitative researcher and software engineer working across market microstructure, applied machine learning and research infrastructure. Corresponding author on a peer-reviewed Q1 finance paper; builds production research platforms that other people rely on. Targeting Computer Science with a quantitative finance focus.

EDUCATION

JBCN International School, Mumbai — Cambridge International (CAIE)

Class of 2027

A Level: Mathematics, Further Mathematics, Computer Science, Economics, Physics, English General Paper · IGCSE: 93%

- Cambridge Independent Research Report (IPQ 9980), 4,982 words: does high-frequency trading destabilise markets under stress? Full independent research qualification assessed to Cambridge criteria.

RESEARCH

Global Attention Saturation Index (GASI)

Borsa Istanbul Review (Q1) — peer reviewed

Lead / corresponding author

- Measured investor attention thresholds across 22 international equity markets, using Lewbel IV identification to resolve endogeneity between attention and volatility.
- Recovered a structural threshold of $\gamma^* = 1.5625$ by profile likelihood (Hansen set [1.14, 2.50]), with Driscoll–Kraay standard errors for cross-sectional dependence.
- Documented the "Silence Signature" — volatility suppression at peak attention — confirmed in 14 of 22 markets under synthetic-control falsification. Passed desk review and a full peer-review round.

Behavioral Ownership Gap in Fantasy Premier League

Journal of Sports Economics (Q1) — under review

Corresponding author

- Formalised the divergence between optimal and actual ownership in a £7.5B ecosystem; Fama–MacBeth panel over 380 gameweeks returned $t = -57.20$.
- Quantified a £137.1M annual welfare cost from information-architecture failure; a pre-registered frozen out-of-sample model returned 105 profitable weeks out of 105 in a long–short replication.

PhenoSync — phenological mismatch and climate value-at-risk

Sole author · Oxford Saïd Climate Challenge

- Built an open-data pipeline over GBIF, NASA MODIS (LST/NDVI), NOAA GHCND and FAOSTAT; OLS shift of 4.57 days/year ($R^2 = 0.83$, $p = 0.0003$) across a seven-framework robustness battery.
- Designed a PMI-severity crop exposure model translating mismatch into billion-dollar value-at-risk indicators.

Why Overparameterised Neural Networks Generalise

Independent exposition

- Derived the Neural Tangent Kernel from first principles using A-level Further Mathematics; argued spectral bias as the resolution of the generalisation paradox, with a fully worked two-neuron example.

ENGINEERING

ATLAS (QUANT360) — quantitative research platform

Solo build · in production use

- Eight institutional-grade modules (Regime, Research, Screener, ML, Volatility, Factor, PCA, Portfolio) built from scratch and used by a CS graduate for real analysis.
- Causal engines for Callaway–Sant’Anna DiD, RD, IV and event studies; volatility suite spanning HAR-RV, the GARCH family, statistical jump models and Diebold–Yilmaz connectedness.
- ML stack with combinatorial purged cross-validation, deflated Sharpe, Double ML, meta-labelling and hierarchical risk parity — entirely on free data (yfinance, FRED, Ken French), with survivorship-bias disclosure.

NEXUS Exchange — agent-based market simulator

Developer / researcher · SSRN paper

- Nine behaviourally distinct trader archetypes over 10,200 Monte Carlo runs (Numba JIT + multiprocessing); recovered $\gamma^* = 1.5625$ by Simulated Method of Moments, independently corroborating GASI.
- Validated all nine canonical stylised facts under jump-filtered ARCH-LM testing; five-tier architecture including a PPO-trained adversarial market maker and an XGBoost/LSTM fragility predictor behind a FastAPI server.

Glassbox SRE — autonomous incident response

Developer

- LangGraph investigation triggered by Prometheus/Alertmanager: parallel commit correlation, pgvector runbook RAG and Prometheus-sourced impact estimation, synthesised into an evidence-cited brief.

- Built a 15-scenario evaluation harness with ground-truth seeded commits: 86.7% top-1 accuracy against a 13.3% deterministic baseline. Figures are queried, never generated.

Daedalus — engineering decision platform

Solo developer · used in STEM Racing competition

- Unit- and dimension-validated execution engine with provenance tracking, Monte Carlo sensitivity, multi-objective Pareto optimisation and equation traceability (Next.js, FastAPI, PostgreSQL, Three.js).

Zeus — agentic execution pipeline

Developer

- Goal decomposition with importance-weighted memory decay, DAG execution visualisation and Playwright automation; end-to-end prompt-to-verified-task loop at 35% lower token cost.

LEADERSHIP & EXPERIENCE

MalkansView — market research firm

Intern · 3 months, ~35 hrs/week

- Broke down company fundamentals and applied analytical frameworks to live market data, following research through to the firm's investment decisions.

Quantitative Finance Club

Founder · 25–30 members

- Designed a 13-week curriculum on markets, valuation and quantitative methods; led the build of Future Lab, a 150-company simulated market with an intrinsic-value engine and Ornstein–Uhlenbeck price dynamics (future-lab-terminal.vercel.app).

Community service — Rural Education Program & Fund Your Dream

Volunteer teacher · fundraiser

- Taught foundational mathematics and English to 50+ children across rural villages, adapting delivery weekly; raised ■25,000+ for study kits, clothing and footwear.

International Award for Young People (IAYP)

Bronze and Silver Awards

- Silver awarded for community service, football goalkeeping and a 26-week quantitative research skill project (Project Atlas).

TECHNICAL

Languages Python, TypeScript/JavaScript, SQL, R | **Research** econometrics (IV, DiD, RD, event studies), GARCH/HAR volatility modelling, factor construction, causal ML, Monte Carlo and agent-based simulation

Engineering FastAPI, Next.js/React, PostgreSQL, pgvector, LangGraph, Prometheus, Docker, Numba, Playwright, Three.js